Status	Finished
Started	Wednesday, 28 October 2026, 11:36 PM
Completed	Wednesday, 28 October 2026, 11:52 PM
Duration	15 mins 40 secs
Grade	**10.00** out of 15.00 (**66.67**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It complicates the analysis process
b. It makes the analysis slower
c. It allows for quicker identification of duplicates
d. It has no effect on the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **2**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By sorting data in a single pass
b. By comparing elements sequentially
c. By using the bubble-up method
d. By breaking the input into smaller parts and merging them

The correct answer is: By breaking the input into smaller parts and merging them

Question **3**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Merge Sort
b. Quick Sort
c. Linear Search
d. Bubble Sort

The correct answer is: Merge Sort

Question **4**

Complete

Mark 0.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Sorting
b. Searching
c. Merging
d. Rearranging

The correct answer is: Sorting

Question **5**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It avoids the need for recursion
b. It only works on small datasets
c. It simplifies the sorting process by using only one pass
d. It allows for efficient parallel processing and sorting of data

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **6**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
b. A sorting algorithm is stable if it preserves the order of non-duplicate keys
c. A sorting algorithm is stable if it preserves the order of all keys
d. A sorting algorithm is stable if it preserves the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **7**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting is rarely used in programming
b. Sorting decreases the efficiency of selection operations
c. Sorting helps in finding duplicates quickly
d. Sorting makes it harder to search for items

The correct answer is: Sorting helps in finding duplicates quickly

Question **8**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It uses the heap data structure
b. It divides the list into sublists
c. It recursively sorts subproblems
d. It bubbles up the largest element in each pass

The correct answer is: It bubbles up the largest element in each pass

Question **9**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Heap Sort
b. Merge Sort
c. Quick Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **10**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Provides a basis for other algorithms to work efficiently
- b. Makes data harder to manage
- c. Makes code execution slower
- d. Decreases the efficiency of algorithms

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **11**

Complete

Mark 0.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It repeatedly finds the minimum element and moves it to the sorted part of the list
- b. It divides the list into two halves, sorts each half, and then merges them
- c. It builds a sorted array one element at a time
- d. It compares adjacent elements and swaps them if necessary

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **12**

Complete

Mark 0.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It increases the number of comparisons needed
- b. It has no effect on the search operation
- c. It allows for faster searching
- d. It makes the search operation slower

The correct answer is: It allows for faster searching

Question **13**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Divide and conquer
- b. Greedy method
- c. Backtracking
- d. Dynamic programming

The correct answer is:
Divide and conquer

Question **14**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

- a. Quick sort
- b. Insertion sort
- c. Bubble sort
- d. Heap sort

The correct answer is:
Insertion sort

Question **15**

Complete

Mark 0.00 out of 1.00

Which built-in Python function is used to sort data?

- a. order()
- b. sort()
- c. arrange()
- d. sorted()

The correct answer is: sorted()